

Commercial Solutions Opening (CSO)

Automated Tool Control & Inventory Management System

Delaware Army National Guard – Army Aviation Support Facility

1.0 Introduction and Purpose

The Delaware Army National Guard (DE ARNG), Army Aviation Support Facility (AASF), is seeking innovative commercial solutions for an Automated Tool Control and Inventory Management System. This announcement is issued under the authority of a Commercial Solutions Opening (CSO) to competitively acquire a system to apply engineered safety controls for reduced human error in tool accountability.

The primary purpose of the system is to mitigate risk of foreign object damage (FOD) in aircraft. Second, the system should increase tool accountability, decrease tool shrinkage, increase maintenance efficiency, and collect tool usage data. Improved inventory management of tools may decrease corrosion of lesser-used tools and improve schedule management of required equipment calibration.

2.0 Problem Statement

We invite companies to submit solution briefs for innovative products that address this problem statement.

The single greatest threat in aircraft maintenance is foreign object damage (FOD), which poses a direct and catastrophic risk to aircraft and air crews. The root cause of most FOD incidents is human error in tool accountability. Manual or semi-automated processes for tool checkout, inventory, and reporting are time-consuming, error-prone, and create administrative burdens on skilled technicians and supervisors. The DE ARNG AASF requires a solution that engineers out this human error.

A system is needed that provides verifiable certainty all tools are accounted for before an aircraft is returned to service. The current state creates unnecessary risks and operational inefficiencies, such as searching for tools and delays in maintenance operations. The ideal solution will automate the tool accountability lifecycle, from checkout to return, providing unalterable audit trails, and streamlining supervisory oversight.

If the Automated Tool Control System is placed directly on the hangar floor, any electrical components sitting within 18 inches of the floor must be specially rated (for example, explosion-proof or intrinsically safe).

3.0 Desired System Attributes and Capabilities

The Government is interested in commercial solutions that possess key attributes such as those listed below. The following list is intended to guide submissions, not restrict them. Proposed solutions might not be able to address every attribute but should address the core problem statement. Discuss trade-offs between attributes and capabilities, cybersecurity and functionality, and present limitations and future capabilities.

- **Automated Accountability:** The system should be capable of automatically detecting and logging every [small] tool that is removed or returned to its storage unit, in real-time, without requiring manual user actions like scanning barcodes or waving wands. Advise similar solution for larger tools.
- **Total Asset Visibility:** Provide a real-time 100% accurate inventory of all tools within the system.
- **Immediate Discrepancy Alerts:** The system should automatically alert the user (with visual and/or audible cues) and supervisors if a tool is missing, a drawer is left open, or an incorrect item is placed in the unit.
- **Access Control, Secure, Keyless, with backup:** Access to tools should be restricted to authorized personnel via secure, individual electronic credentials (for example, biometrics, personal identification number, proximity card, or integration with Government-furnished credentials like the Common Access Card). There may need to be a non-electrical or non-computer backup at the technician or supervisor level, which would necessitate its own controls.
- **Cybersecurity and Functionality:** Data created and maintained by the systems must be kept confidential, guarded against destruction or modification, and available when needed. Contractor/Manufacturer must disclose what data it collects, if any. Advise protocols and procedures for software bug patching.
- **Unalterable Audit Trail:** The system must create a permanent digital log of every transaction, including user ID, tool(s) removed and replaced, with a precise timestamp. This log must be secure from tampering.
- **Centralized Management & Reporting:** The entire system, particularly if more than one tool storage unit, should have the ability to aggregate and generate reports on tool use, user activity, and inventory status.
- **Scheduled Notifications:** Supervisory personnel should be able to periodically program the system to alert users on a regular or ad hoc basis that select tools need to be inspected or calibrated.
- **Data Portability & Architecture:** The system should be built with non-proprietary interfaces and allow for easy export of data such as inventory and logs into a standard format, for example, CSV or XML. Describe your system architecture, for example, modular open system architecture, proprietary, or hybrid. If hybrid, to what component level. Describe what mating and attachment characteristics permit other vendor modules to interface with your system.
- **Tool Procurement:** DE ARNG may procure individual tools concurrently with the instant procurement of the tool control system, later in Fiscal Year 2026, or beyond. Indicate ability to track and control tools and equipment from vendors other than that of the automated tool control system provider. Provide scenarios to identify and procure mechanics' tools and equipment, with the system or later, and ease with which supervisors can add or remove tools to the tool control system
- **Battery Power and Offline Functionality:** System should be independently powered and portable as it will be moved frequently to vicinity of aircraft being serviced. In event of a network or power failure, the system should continue to authenticate users, issue, and track tools, with data automatically synchronizing when network or power is restored.
- **Terms and Conditions:** Consistent and customary to those provided to commercial customers but none that are prohibited by statute in contracts with the U.S. Government.

4.0 Submission Process

Phase 1. Solution Brief Submissions will be accepted through 31 May 2026.

Interested parties must submit a clear and concise description of their solution. The submittal should include:

- Description of the proposed solution(s) and technical approach, including a discussion of how the solution addresses the [2.0] Problem Statement and [3.0] Desired System Attributes and Capabilities.
- An illustrative contract line item structure for the Government to procure the solution.
- High-level schedule(s) indicating speed of execution versus functionality or capability of deliverables. If solution is delivered in phases, describe phases. Describe lead times and supply chain vulnerabilities.
- A plan for scalability from one system covering one work area to a system(s) covering up to six work areas. Assume DE ARNG procures one system this fiscal year, but DE ARNG could procure all its requirements as early as this fiscal year or over a few fiscal years, as funding permits.
- Software bill of materials, if solution is software intensive.
- Description of electrical or telecommunications hook-ups required of DE ARNG AASF, if any.
- Preliminary list of operation, maintenance, installation, and training data you provide with your solution.
- Description of feedback you would request from DE ARNG AASF to further improve product(s).
- [Optional] Preliminary fixed pricing of the above-requested line items, including extended warranty options, if any. Advise pricing if system and tools are procured simultaneously versus separately. Advise if pricing is on a one-time basis or on a subscription basis. Suggest options and option pricing, if any.
- Where applicable, mark and identify documents being submitted on a proprietary basis.

Phase 2. Demonstration/Pitch/Negotiation (By Invitation)

Based on a review of submittals, the Government may invite select one vendor, multiple vendors, or no vendors to provide a live or virtual demonstration of their solution; and may invite select vendors offering solutions most advantageous to the Government for negotiation. Negotiations may occur concurrently with—or subsequent to—the demonstration. Negotiations may be conducted during the submission acceptance period.

Phase 3. Evaluation and Award

USPFO Delaware will select one solution provider and issue a fixed-price FAR-based purchase order (SF 1449 or DD 1155). The awardee should expect a typical Government purchase order with Revolutionary FAR Overhaul contract clauses requiring bilateral or counterpart signatures. Operation & Maintenance funds are currently available to award a reasonably priced contract this fiscal year. Target award date is no later than 31 Jul 2026.

Phase 4. Post-Award

System sustainment, administration, and evaluation of solution impact on warfighter support. Option exercises, if any, if advantageous to the Government.

5.0 Evaluation Factors

Solution submittals will be evaluated based on the offeror's ability to provide a compelling, effective, and feasible solution to the problem. Evaluation factors are listed in order of importance:

1. **Technical Merit:** The degree to which the proposed solution effectively addresses the problem of automated tool accountability and FOD prevention.
2. **Operational Impact:** The potential for the solution to increase productivity, reduce operational risk, and either integrate into existing maintenance workflows or readily transform our maintenance process.
3. **Usability and Scalability:** Adherence of the solution to open or voluntary consensus standards and data portability. Or reasonable license rights to proprietary standards customarily available to commercial users.
4. **Company and System Viability:** The company's experience and ability to deliver and support the proposed commercial solution for the life of the product, including customary warranties. Past performance applies.
5. **Compliance:** [RFO] FAR/DFARS Part 25 Foreign Acquisition / Buy American and FAR/DFARS Part 40 Information Security and Supply Chain Security.
6. **Pricing:** Acquisition cost and life-cycle cost; fair and reasonable.

6.0 Acronyms and Applicable Publications

AASF	Army Aviation Support Facility
ARNG	Army National Guard
BAA	Buy American Act
C-SCRM	Cybersecurity Supply Chain Risk Management
COO	Country of Origin
CSO	Commercial Solutions Opening
DA PAM	Department of Army Pamphlet
DD/DoD/DoW	Department of Defense / Department of War
DE	Delaware
DFARS	Defense Federal Acquisition Regulation Supplement
FAR	Federal Acquisition Regulation
FOD	Foreign object damage
OMIT	Operation, Maintenance, Installation, and Training
RFO	Revolutionary FAR Overhaul
SF	[GSA] Standard Form
USPFO	U.S. Property & Fiscal Office (Contracting Authority)

DA PAM 385-10 Army Safety and Occupational Health Program Procedures, dated 24 Jul 2023, Paragraph 6-3 Foreign Object Damage (Advisory) ... <https://armypubs.army.mil/ProductMaps/PubForm/PAM.aspx>

DA PAM 750-43 Army Test, Measurement, and Diagnostic Equipment Implementation Procedures, dated 15 Jul 2024, Paragraph 7-11 Calibration Interval (Advisory) ... <https://armypubs.army.mil/>

NFPA 70 National Electrical Code, Article 513 Aircraft Hangars